Question ID: 628300a9

Question

A science teacher is preparing the 5 stations of a science laboratory. Each station will have either Experiment A materials or Experiment B materials, but not both. Experiment A requires 6 teaspoons of salt, and Experiment B requires 4 teaspoons of salt. If x is the number of stations that will be set up for Experiment A and the remaining stations will be set up for Experiment B, which of the following expressions represents the total number of teaspoons of salt required?

Answer

A. $5x$

B. $10x$

C. $2x + 20$

D. $10x + 20$

Question ID: f5ff91b2

Question

If $\frac{x-5}{7} = \frac{x-5}{9}$, the value of $x - 5$ is between which of the following pairs of values?

Answer

A. -9 and -7

B. -3 and 3

C. 4.5 and 5.5

D. 6.75 and 9.25

Question ID: 429fb7c0

Question

What value of t is the solution to the equation $0.8t - 0.46 = 8(t - 0.001) + 1.9$?

Question ID: 3f8a701b

Question

The equation $9x+5=a(x+b)$, where a and b are constants, has no solutions. Which of the following must be true?

I. $a=9$

II. $b=5$

III. $b \neq \frac{5}{9}$

Answer

A. None

B. I only

C. I and II only

D. I and III only

Question ID: aee9fd2d

Question

If $\frac{x+6}{3} = \frac{x+6}{13}$, the value of $x + 6$ is between which of the following pairs of values?

Answer

A. -7 and -3

B. -2 and 2

C. 2 and 7

D. 8 and 13

Question ID: 771bd0ca

Question

$$5(t+3)-7(t+3)=38$$

What value of t is the solution to the given equation?

Question ID: e6cb2402

Question

$$3\left(kx+13\right)=\frac{48}{17}x+36$$

In the given equation, k is a constant. The equation has no solution. What is the value of k?

Question ID: 25e1cfed

Question

How many solutions does the equation $10(15x - 9) = -15(6 - 10x)$ have?

Answer

A. Exactly one

B. Exactly two

C. Infinitely many

D. Zero

Question ID: b7e6394d

Question

Alan drives an average of 100 miles each week. His car can travel an average of 25 miles per gallon of gasoline. Alan would like to reduce his weekly expenditure on gasoline by $5. Assuming gasoline costs $4 per gallon, which equation can Alan use to determine how many fewer average miles, m, he should drive each week?

Answer

A. $\frac{25}{4}m = 95$

B. $\frac{25}{4}m = 5$

C. $\frac{4}{25}m = 95$

D. $\frac{4}{25}m = 5$

Question ID: f14484a5

Question

A manufacturing plant makes 10-inch, 9-inch, and 7-inch frying pans. During a certain day, the number of 10-inch frying pans that the manufacturing plant makes is 4 times the number n of 9-inch frying pans it makes, and the number of 7-inch frying pans it makes is 10. During this day, the manufacturing plant makes 100 frying pans total. Which equation represents this situation?

Answer

A. $10(4n) + 9n + 7(10) = 100$

B. $10n + 9n + 7n = 100$

C. $4n + 10 = 100$

D. $5n + 10 = 100$

Question ID: 2937ef4f

Question

Hector used a tool called an auger to remove corn from a storage bin at a constant rate. The bin contained 24,000 bushels of corn when Hector began to use the auger. After 5 hours of using the auger, 19,350 bushels of corn remained in the bin. If the auger continues to remove corn at this rate, what is the total number of hours Hector will have been using the auger when 12,840 bushels of corn remain in the bin?

Answer

A. 3

B. 7

C. 8

D. 12

Question ID: ac472881

Question

$$\frac{12x+28}{4} - \frac{s}{13} = r(x-8)$$

In the given equation, s and r are constants, and $s > 0$. If the equation has infinitely many solutions, what is the value of s?